

Topic 29 -- Intro to Differential Equations

Some Definitions

A differential equation is an equation that contains an unknown function and some of its derivatives.

The order of a differential equation is the order of the highest derivative that occurs in the equation.

For this introduction, we will only consider “ordinary” differential equations (i.e. equations that contain ordinary derivatives).

Example 1

For each equation, determine the order and the implied dependent and independent variables.

$$y' = 1 + 7x - 4y \qquad y'' + y' + 12y = 0$$

$$\frac{dy}{dt} = y^4 - 6y^3 + 8y^2 \qquad \ddot{x} + 4x = \sin^2 2t$$

Definition of solution

A function f is called a solution of a differential equation if the equation is satisfied when $y = f(x)$ and its derivatives are substituted into the equation.

Example 2

Show that every member of the family of functions $y = Ce^{x^2/2}$ is a solution of $y' = xy$

Definition of Initial Value Problem

An Initial Value Problem (IVP) is a differential equation combined with some initial value(s) of the dependent variable.

Example 3

Solve $y' = xy$, $y(0) = 5$

Example 4

Determine the values of the parameter r such that $y = e^{rx}$ is a solution of the equation $y'' - 3y' - 18y = 0$.

Show that $y = c_1y_1 + c_2y_2$ is a solution of the differential equation.

Solve: $y'' - 3y' - 18y = 0, y(0) = 1, y'(0) = -4$

Example 5

For what nonzero values of k does $y = \sin(kt)$ satisfy the differential equation $y'' + 4y = 0$?

Can you think of another function that will solve $y'' + 4y = 0$? Verify!

Show that $y = c_1y_1 + c_2y_2$ is a solution of the differential equation.

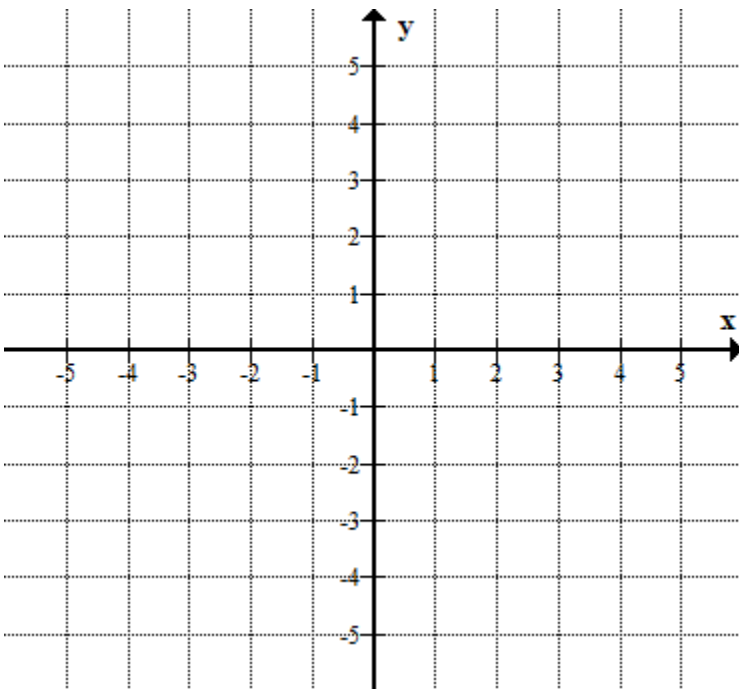
Solve $y'' + 4y = 0, y(0) = 2, y\left(\frac{\pi}{4}\right) = 3$

Example 6: Slope Fields

Consider the initial value problem: $\frac{dy}{dx} = xy, y(1) = 2$

We do not know (yet) how to determine the solution curve for the equation. Can we visualize a solution curve?

Let's create a slope field for the equation!

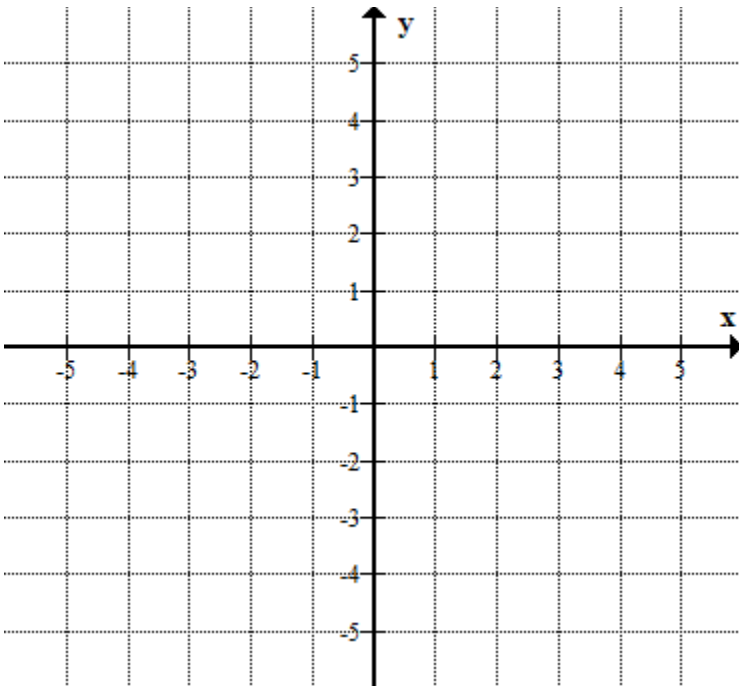


Example 7

Create a slope field for the equation $y' = \frac{x}{y}$.

Then draw possible solution curves for the initial value problems:

- $y' = \frac{x}{y}, y(1) = 2$
- $y' = \frac{x}{y}, y(-3) = 0.5$



Example 8

A function $y(t)$ satisfies the differential equation

$$\frac{dy}{dt} = y^4 - 6y^3 + 5y^2$$

- What are the constant solutions of the equation?
- For what values of y is y increasing?
- For what values of y is y decreasing?
- Draw a possible solution curve for $\frac{dy}{dt} = y^4 - 6y^3 + 5y^2, y(0) = 3$

Introduction to Modeling

When a physical situation includes a rate of change, it may be modeled with a differential equation.

Example 9

“Suppose a population grows at a rate proportional to its size.”

Example 10

“Suppose the rate at which a rumor is spreading in a fixed population is proportional to the product of the number of people that have heard the rumor and the number of people that have not heard the rumor.”